3D Environments with Games Characteristics for Teaching History: The VRLerna Case Study

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ABSTRACT

The latest rapid advancement of technology of informatics and communication brought new prospective to education. However, the potential of those technologies could have been exploited further in order to transform students from passive data receivers, to active actors in the learning process. The scope of this paper is the presentation of a theoretical educational framework for the use of interactive three dimensional virtual environments with game characteristics that aims at the initiation of motivation and enhancement of learning. To evaluate the effectiveness of the theoretical framework a prototype educational application called “VRLerna” has been implemented and evaluated with real users. VRLerna is an interactive reconstruction of the central building of a prehistoric population at Lerna of Argolis which is used to teach history at Key stage 5. The paper describes the methodology used to evaluate the prototype and via this the theoretical educational framework and presents the evaluation results that portray the success of the prototype to attract the students interest and keep them engaged and motivated.

Categories and Subject Descriptors

K.3.2 [**Computers and Education**]: Computer and Information Science Education – *Computer science education*

General Terms

Theory

Keywords

History education, virtual environments, video games, constructivism, motivation.

1. INTRODUCTION

The most wide spread theories of learning psychology suggest that the learner’s activation and engagement in an energetic learning process are prerequisites for achieving certain learning goals. According to Rogers active learning is linked with motivation [55].

At the same time, modern history teaching considers the ability of being critical on all historical narratives [38] and encounters the understanding of time, change or continuity [35] as very important educational goals. It is therefore broadly acknowledged that teaching facts through textbooks is no longer enough in order to appreciate the past. As a result, archaeological evidence seems to become more valuable under the new circumstances [27] [59] [3] [45]. Hamilakis (2004) suggests that archaeology in education can help students develop critical, as well as synthetic, analytical and hypothetical abilities as they come to understand the dialectic relation between the past and present through the study of material culture. In order to address those goals motivating, challenging and interactive means that avoid one-dimensional narrative [62] and can support the students to gain a non-fragmented view of the past are required. Those are pursued in modern technology which provides the tools for reconstructing what time has destroyed and offers the environments in which the learner can engage a rich empirical experience [16].

As a result, it is considered necessary to enhance the existing learning process by adding elements that aid into motivating and engaging the students. Learning technology can play an important role in achieving this. More specifically, technologies such as Virtual Reality (VR) and video games could be used as learning tools in order to create the context into which the students could remain active through entertainment, commitment and motivation.

VR is a technological discovery which allowed to overcome the classical way of interaction with personal computers through static means of interface such as the mouse and the keyboard and step through the computer screen into a three dimensional (3D) artificial world. It is a multi-sensory experience, based on: real-time 3D computer graphics; stereoscopic rendering; head/eye/body tracking; and binaural vision [22]. In a virtual environment (VE) the users with the help of suitable devices has the sense of immersion in an artificial world in which they have the sense of moving and interacting with its content in a intuitive and natural way [10], while the reproduced environment changes reflecting the user’s movements.

VR offers the potential to create pioneering educational environments [43] [44] and the ability to explore and interact with objects and the environment [42]. It uses metaphors borrowed from the games and the theater, promoting sentimental elements in the interface [37]. Many scholars and teachers believe that VR offers many advantages which can support the learning process [49]. For some, these advantages focus on its ability to facilitate the use of constructivist learning activities [68]. For others, it is the fact that it offers alternative ways of learning which can help different types of students, as for example optical ones. Until today there has been many examples of the use of VR for educational purposes and subjects [48], including those of history [57] [46], as it has all those characteristics which allow the "revival" of the past (archaeological finds and sites) and the study of historical issues.

Nevertheless, a common problem experienced in VEs is the lack of structured activities and processes [19] [31]. This is because the initiative of exploration in the VE moves to the users themselves. Another problem focuses on the lack of 3D VEs to offer the necessary dynamics in order to provide a rich and focused interaction, which can trigger the student's interest and motivate them to fulfill certain learning goals.

At the same time, video games technology is a rapidly evolving technological field. It is considered a technology which penetrates more and more in modern societies and is becoming one of the main, most lucrative and effective forms of entertainment, maybe the most popular among young people [2] [33]. Many scholars support the use of video games as a means that aids learning. This opinion is based on the enormous influence of games upon young people as well as on the fact that they seem to motivate students with a more constructive way than the one used by conventional learning methods [36] [54] [20] [34]. Boyle (1997) considers that games are able to give an attractive and pleasant form to learning, by offering a strong "pattern" on which we can design effective learning environments. Papert (1993) also supports that video games give a vivid and interesting pace to teaching, in regards to conventional means used in most schools, which turn teaching into a slow and unattractive process [65]. Modern students are familiar with video games technology and are most likely to enjoy a learning experience incorporated in a video game [5] [54].

An educational video game should accomplish two goals: entertainment and learning. Both goals should be accomplished in a rather satisfying level in order to achieve an effective educational environment.

Entertainment is very important and strongly connected with student motivation [25] [60] [2]. Successful entertaining educational video games are considered the ones that offer intriguing challenges, have a clear goal and specific outcome, as well as precise, constructive and encouraging feedback and offer elements such as those of curiosity and imagination [40]. Prensky [53] suggests that educational games which lack entertainment disorientate and offer nothing more than conventional teaching software. In addition, learning is achieved only in an environment which offers the necessary "teaching support", meaning adopting such pedagogical and teaching principles (learning theories, educational methods, and learning tools) which are consistent with the video games' character and can lead to achieving educational goals [54].

Nevertheless, through literature review of prior attempts to develop educational video games, it is being realized that the achievement of a balance between entertainment and learning is complicated. This is the reason why there are few successful educational games [12]. Indicative of the problem is Brody's [9] opinion, who mentions that the attempts to combine educational content and entertaining elements have produced some unsuccessful educational video games and some ineffective entertaining learning experiences. As a result, there is a need to design and develop educational games which will combine stimulating elements with clear learning goals and a satisfying educational content.

The main purpose of this paper is to present a theoretical framework for designing and developing educational 3D VEs that reflect the main characteristics of video games (see Section 2). In order to evaluate the effectiveness or this theoretical framework a pilot educational application is implemented, the "VRLerna" (see Section 3). Section 4, presents the methodology that has been used to evaluate the educational outcome, while section 5 presents and discusses the evaluation results. Finally, the paper closes with conclusions and future recommendations (see Section 6).

2. EDUCATIONAL FRAMEWORK

The basic goal of the theoretical framework discussed in this paper is the reinforcement of learning results by initiating motivation. However, in order to accomplish this goal it is essential to understand the main factors that motivate learner and what is the desirable learning outcome. The desirable outcome includes the focus of learners' attention, the degree of their excitement and involvement in the learning process and insisting on achieving the learning goals [13] [51]. Moreover, motivated learners should feel enthusiastic, focused and engaged by their goal. Their interest for learning should be evident and the learning process should be pleasing.

The discussed theoretical framework consists of two sections: the educational framework to be satisfied and the educational tools to be used to achieve this (see Figure 1). The educational framework consists of the pedagogical background which is being adopted. The later includes learning theories and educational methods which help keeping the learners engaged and motivated and addresses the characteristics of the chosen educational tools. More specifically, it includes the constructive learning theory, which focuses on the learner, who is called to act in order to obtain knowledge [58] [56] [18] [17] [28]. The educational methods that have been adopted as part of the educational framework are those of experiential and discovering learning. Those address the basic goals of the theoretical framework, which is to motivate the learner, as well as the basic principles of constructive learning theory, as they give an impulse to learning through experience and personal knowledge discovery.

The educational tools that have been adopted are based on 3D VEs and video games, that can enhance the learning process, as stated earlier (see Section 1), if the educational dynamic of certain characteristics is triggered, like those of: fantasy; curiosity; rules/goals; control; and as partial but equally important those of interaction (piloting, choosing objects etc); realism (graphics, sound); and the video game's environment. The activation of such educational characteristics can be realized through integrating those video games' factures in a certain educational framework [21]. As a result, the elements that have been chosen are consistent with basic principles of the constructive learning theory, the educational methods of experiential and discovering learning and those which have the ability to engage and motivate the learner. In addition, the elements that have been chosen can be addressed in a 3D VE.

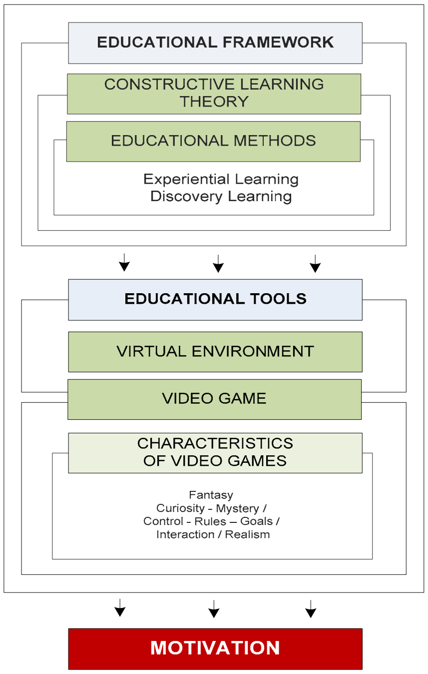

Figure 1 The structure of the theoretical framework for the reinforcement of learning by initiating motivation.

3. VRLerna case study

The basic elements of the theoretical framework stated earlier (see Section 2) have been applied in the "VRLerna" pilot application. The VRLerna pilot is an interactive 3D VE with educational goals and the basic characteristics of a video game.

This project is based on the reconstruction of the "House of Tiles" in Lerna, an archaeological site with a particularly long history, reflected in the Greek mythological tradition. The edifice is dated in the Early Helladic Period (3000- 2000 b.C.) [11]. One of the reasons for choosing this site for reconstruction is the presence of thorough archaeological publication related to it, which provides the required data for its reconstruction. Another reason for choosing to work around this site is that it can help to cover the great economic and social changes which led to the development of the hierarchic society in the Early Helladic Period, a subject of great importance which however, it is not covered by the Greek history curriculum. Last but not least, the building itself is of immense importance due to its impressive dimensions, its elaborate construction and its rare architectural type [23] (what is most important is its function, as it has been considered by many scholars as the house of certain members of the local elite, a community center, a center for exchanging commodities, or a site for the redistribution of community wealth) [65].

The main educational goal of the current project is to teach certain aspects of the ancient Greek history in the secondary education using methods that reinforce structure and facilitate learning. More specifically, the current project aims at presenting elements that are not covered in the ordinary teaching process. Moreover, it is believed to assist the students to have a first acquaintance with the Early Helladic Period in the Greek mainland in order to acquire basic skills in making interpretations about the period and associating it with latter social developments.

The structure of VRLerna consists of three different levels: the operational level; the multimedia level and the VR level. During the execution of basic functions using VRLearna those three levels either interact with each other, or they function separately. The operational level is the "brain" of the system, defining how and when the two other levels will function. The most important element of the operational level is a data base. This data base manages user actions (a table where the actions of the user and his answers to the knowledge questions are being stored), objects (a table where the objects – archaeological finds that are chosen and observed by the user are being stored) and VE (where the route followed by the user is being stored). The multimedia level consists of the application graphics, as well as applications that store instructions for the game, explanations about the game's goals, information about the historical background of the "House of Tiles", and user feedback according to their actions are presented. The VR level consists of all the VEs and 3D objects of VRLerna.

The implementation of VRLerna consists of the 3D reconstruction of the "House of Tiles", the application interface and finally the unification of all the structural elements in a common interactional environment. The 3D reconstruction was a rather difficult and time consuming process, as a thorough study of the bibliography (the publication, articles, previous reconstructions and photos of the archaeological site and its artifacts) was required. For building the three dimensional models a 3D modeling software has been used (3D Studio MAX of Autodesk). For programming the basic interactions of the VE (navigation in the VE, basic interaction in the VE and objects in it) Virtual Reality Modelling Language (VRML) has been used. Adobe Photoshop and Flash software were used for creation two dimensional graphics and presenting the informative material respectively. Finally, for bringing all the elements together in a unified interactive environment HTML, PHP and mysql programming languages have been used.

Figure 2. 3D recreation of the "House of Tiles" in Lerna, Argolis dated in the Early Helladic Period (3000- 2000 b.C.)

VRLerna has been designed and implemented addressing the principles of the constructive learning theory. The learners take active part in the educational process as they explore the rooms of the House of Tiles, discovering archaeological findings and obtaining new knowledge. The learners interact almost naturally with the environment in a way which simulates everyday activities. As a result, knowledge for the life in this specific historical period is obtained through their own experiences and not through narratives and descriptions [67].

The experience of interaction with the VRLerna application incorporates elements which trigger the imagination and the curiosity, enhance the mystery atmosphere and have the potential to keep the learners focused and interested. The realistic VE is rather satisfying as it offers the ability to get oriented in the space and adjust to it. The interaction (navigation, choosing and manipulating objects) was designed on well known forms/ frames so that the user would not be disorientated.

The 3D recreation of the space offers new potential of visualisation and interaction with the educational material by using basic pedagogical principles [8].

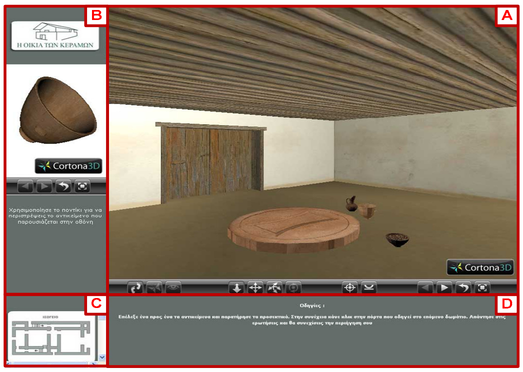

Figure 3. Snapshot of the basic interface of VRLerna. The interface consists of the four following areas: (a) depicts the 3D reconstruction of the environment; (b) is a window that allows the user to view a selected object closer ; (c) is a top view of the reconstruction that shows the current position of the user; and (d) is a dialog box were basic directions to the user are provided.

Realism (geometrical models, textures and lighting) is rather satisfying as it offers the ability to get oriented in the VE and get adjusted to it.

Navigation in the VE has been designed to assist the users not to get disoriented and remain focused on the important activities taking place in the VE [14] [19]. It adopts the natural movement of walking, one of the most efficient movements in VE, as it simulates a user's everyday life experience [61]. The user viewpoint follows the self-centred frame, according to which the user observes the space in a way similar to the one they do in real world. The navigation as well as the interaction with objects in the VE takes place by using the mouse and the direction arrows of the keyboard. It is an interaction which is based on the users' previous knowledge and so it facilitates their orientation in the VE [47].

The navigation in the application triggers the learners' imagination. The learners come in touch with the past which is not part of their conventional everyday life and cannot be approached otherwise. Previous researches claim that the incorporation of imagination elements in the educational environment can cause greater interest and advanced learning outcome [13] [51].

When entering the VE a dialogue box appears which informs the learners about the game's basic goal which is tracing archaeological finds and the ways to achieve this. Next, a short presentation enriched with text, diagrams and photos provides information about the historical period the settlement and the building. The information is clear, simple and without lengthy details. The clear presentation of the goals and the rules before engaging with the game's activities contributes in motivating the learner. Related research has shown that sparing information [40], surprise, unexpected outcomes [6] and the inability to predict future developments [30], enhance the sense of mystery. As mystery elements can be considered the incorporation of activities in a context that enforces imagination [4], the quest for archaeological finds and the exploring of unfamiliar spaces of the building. Also, Malone and Lepper [40] consider that mystery triggers curiosity which is one of the main factors that lead to learning.

The learners have to explore and access all the levels of the game (one in the house yard, six in the basement and one on the floor). In each one of them they can see various artefacts which are three dimensional reconstructions of archaeological finds from the Lerna excavations. The learners are called to trace those (using the game instructions). By selecting an artifact in the VE, a 3D reconstruction of it appears in a separate window at the left side of the screen, where it can be rotated and observed closer (see figure 3). This feature is very important, as according to the constructivist learning theory, the ability to observe an object under many and different sights contribute to making an issue simpler and easier to understand [29].

In order to move to the next level the learners have to complete a quiz assessing their knowledge gained by playing the game (see Figure 4). The quiz questions concern the relevant level of the game. The quiz questions are of increasing difficulty in order to enhance motivation and learner's performance [39]. Malone and Lepper [40] claim that players are challenged by quizzes of medium difficulty. In those cases where the difficulty level is either too low or too high according to the learners' abilities, the player/learner's interest is getting lost soon enough and any effort is being abandoned respectively [52]. After the learners have gone through the knowledge assessment quiz they receive feedback about their achievement. If the users answer a question correctly they continue to the next level. If they fail answering correctly they have the chance to try again. User feedback during the game motivates and prompts them to readjust their behavior according to their performance [32]. During the exploration process the system assists the learners by providing short instructions regarding the directions they have to follow and the action they have to take every time.

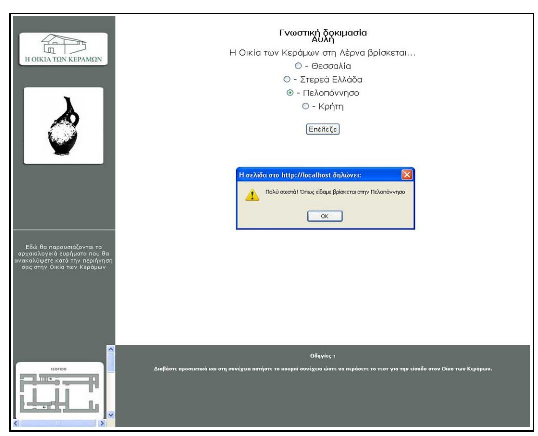

Figure 4. A snapshot of the quiz the learners have to complete after they have navigated through a level of the VRLerna game and they have gone through the objects contained in it.

In order for the application to function a PC is required with a minimum 2GHz processor, a 1GB RAM memory, and a 1024x768 screen analysis. As to the required software a network explorer must be available (Mozilla Firefox 2.0 or latter, Internet Explorer 5.5 or later, Netscape Navigator 6.0 or later) which can support the HTML 4.0 and CSS models. It is also considered necessary to have installed programs that allow the projection of VRML (wrl) and Adobe Flash (swf) files.

4. EVALUATION

The VRLerna pilot application was evaluated in order to:

- assess its efficiency as a learning tool
- examine its dynamics to motivate the learner
- study the effectiveness of the theoretical framework.

The evaluation took place among students of the 3rd high school grade class (Key Stage 5 students, 16–18 years old) in a small town in central Greece. The particular class has been chosen based on the fact that in the past they have been involved in activities related to Early Helladic Period. Seventeen students aged 14-15 years old, eight of which were girls and nine boys took place at the study. The sample size was determined by the number of the students in the 3rd grade class.

During the preliminary stage of the evaluation the principle and all the involved teachers and students were contacted and informed about the aims and the goals of the research in order to obtain their active participation and provide assurance of the confidentiality of the personal data.

The method chosen for the evaluation of the educational efficiency of the interventionist method (interaction with the pilot application) is that of the pre and post-test. According to this the data collected assessing the students' knowledge before and after their interaction with the educational application is compared.

The evaluation study took place in two phases. The first one took place inside the ordinary school class where a twenty five minutes lecture was given covering themes related to the historical topic dealt with at the VRLerna pilot, followed by a discussion. Next, the students had to complete a questionnaire or the pre-test, which apart from specifying the sex it comprised of ten "closed" statistical questions. Seven of those were related to the educational content of the application and aimed to explore the students' previous knowledge. The other three concerned the learners' attitude towards personal computers, their experience with using them as educational aids and their familiarisation with video games. It should be noted that in order to ensure anonymity and be able to compare results in pre-test and post-test, no names were used, instead a unique number – code for each questionnaire was used.

In the second phase of the study the students interacted with VRLerna in the school's computer lab. The students shared one PC in pairs. At the beginning the students were asked to use an experimental 3D VE, which gave them the ability to get familiar with navigating in a VE. Before entering the VRLerna the students had the chance to interact with an educational multimedia application that covered the theory they have been taught during the first phase of the study. Next, the students were left alone to interact with VRLerna for a typical teaching hour (forty-five minutes). The main activities to take place were given to the students by the system. The researchers supervised the process and interfered to resolve problems, mostly technical ones.

After the end of the interaction with VRLerna the students were asked to complete a post-test questionnaire, that was also anonymous and the relation with the pre-test questionnaire was ensured with the aforementioned unique code. The post-test questionnaire consisted of two parts. The first part similarly with the pre-test questionnaire consisted of seven multiple choice questions assessing the learning outcome. The second part consisted of 19 questions (16 "closed ones" and 3 "open") that targeted at ascertaining directly and in-directly if the application managed to keep the students motivated throughout the interaction with it. More specifically, the questions chosen aimed in finding out if the specifications laid by the structural elements of the theoretical framework caused the expected learning results (motivation, learning reinforcement).

The main keystones of the study revolve around the pedagogical framework and the educational tools. As a result, the questionnaire comprises questions aimed at extracting knowledge related to the application features to support imagination, the sense of mystery and curiosity, the way that the game's goals and rules are presented to the learner, the efficiency of interaction between the learner and the system and elements that enhance realism in the VE. Also, there were questions assessing the constructive approach the application adopted for presenting educational material, as well as those application features to motivate the students and how this affected the learning outcome. The questionnaire includes also some open questions allowing the students to express their likes and dislikes and what they would like to see being added to the application.

5. RESULTS

This section presents the evaluation results of the study described above (see Section 4) are presented. Figure 5 below presents the results that derived from the pre-test and post-test comparison, in relation to the learning potential of the interventionist method that have been used. The results indicate that there has been an impressive change in the number of students that answered correctly after using the pilot application. More specifically, the mean of the correct answers given by the students rose from 7,57 during the pre-test to 13,43 after the post-test. In order to compare the learning performance before and after the interaction with the pilot application the Wilcoxon criterion has been used, which is considered suitable when the sample is small and there are serious doubts about the regularity of the population [66] [68] [69]. The results showed that the raise in the learning efficiency was statistically important (Z=--2,226, p=0,026).

One of the basic aims of the present study is to motivate the learners. Motivation helps retaining the learners' attention and focus until the expected learning goals have been fulfilled [1]. The results that derived from the post-test study showed that the application's environment triggered in a certain degree characteristics such as, the imagination and mystery, which indirectly affected the learner's motivation (Figure 6). More specifically, it appears that when the learning content is characterized by elements of imagination it attracts more the learners, it causes greater interest and it contributes to advanced learning results [13] [51] [1] [52]. Many students that interacted with the application (35%) consider that the environment contributed much or very much in triggering their imagination.

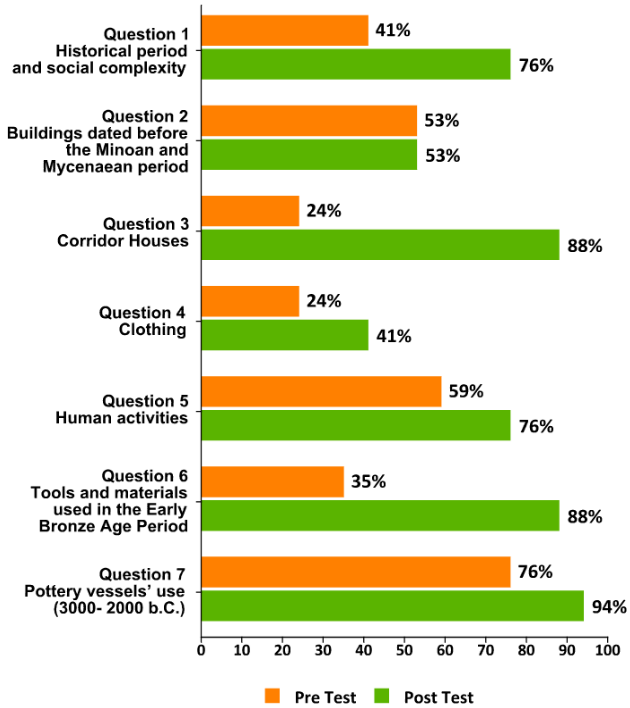

Figure 5. A comparison of the percentage of students' correct answers on topics that have been covered during pre and post-test

However, taking in mind the percentage of those who answered little (48%) and not at all (18%), it is considered that further activities that trigger the learners' imagination should be included. As to the elements of mystery and curiosity, Malone and Lepper [40], suggest that they can become important factors in assisting learning, if they are included in the learning process. The answers of the students during the evaluation prove that the above characteristic exists in VRLerna in some degree, as the 47% of the sample mentioned that the game's action created a mysterious atmosphere that generates curiosity. At this point, it should be mentioned that the due to their subjective nature imagination and mystery are difficult to be measured and lead to objective conclusions.

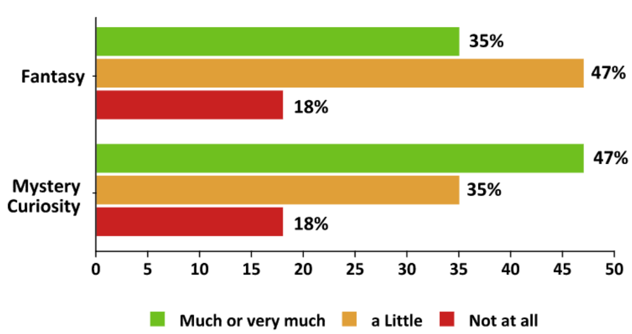

Figure 6. The graph depicts the students' response assessing how much the VRLerna environments stimulated their fantasy – mystery/curiosity.

Significant are the results regarding stimulating motivation, like attracting the learners' interest and entertainment (see Figure 7). The evaluation results showed that 64% of the students found the experience of interacting with the pilot application as very, or extremely interesting. This result is very important since related bibliography mentions that interest, motivation and learning are directly connected [15] [50] [4] [63].

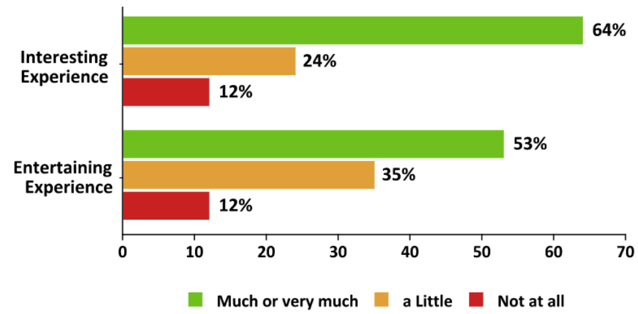

Figure 7. The graph depicts the students' response assessing if they found the VRLerna game interesting and entertaining.

Also, the entertaining factor of video games is of major importance since its existence or absence is directly linked with learners' engagement and consequently their motivation [25] [60] [13] [2] [54]. Malone [41] consents with this opinion and suggests that successful educational games are entertaining and as a result motivating. The results showed that 53% of the students considered that their experience interacting with the educational VE was very, or extremely entertaining.

Finally, the results regarding the reinforcement of the learning outcome (see Figure 8) showed that 59% of the students using VRLerna considered it contributed much or very much in understanding the educational content and more specifically the prehistoric period in Greece. Results showed that the ability to present an artifact in a different window and providing the opportunity to rotate and manipulate it was very helpful. This way of presentation and interaction with objects follows one of the basic principles of the constructivist theory, according to which an object observation via different viewpoints contributes to simplification and easiest appreciation of it [28]. The above opinion is being confirmed by the research results according to which the 64% of the students considered that this ability contributed much, or very much to the careful observation of the archaeological finds.

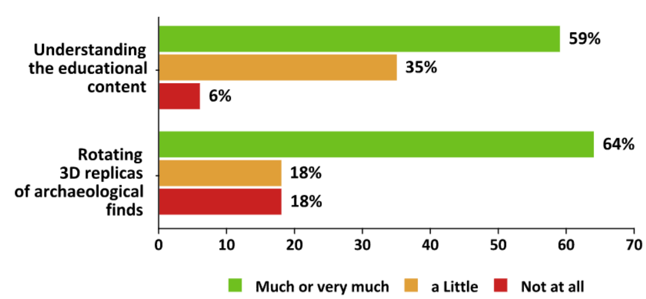

Figure 8. The graph depicts of the students' response assessing if the use of VRLerna environment contributed to their understanding of the educational content and how helpful they found the feature of having a closer look and rotation an object.

6. CONCLUSIONS

The previous sections presented the design, the realisation and the evaluation of the pilot application VRLerna. It is an interactive 3D VE which holds the main characteristics of a video game. Its educational content aims to facilitate the teaching of historical/ archaeological issues and it addresses to Secondary Education students (high school). The application's realisation follows a series of specifications which are part of a theoretical framework which aims at stimulating motivation and improving the learning efficiency. The analysis of the data which were gathered during the evaluation of the application with real users showed that the pilot application contributed in a significant degree in improving the learning efficiency of the students. It was also established that the theoretical framework on which the application is based can trigger the interest, entertain and motivate satisfactorily the learners. Nevertheless, there is a need for further improvements as failings and weaknesses have been noted. More specifically, as it has been realised by answering the questionnaires used at this study, realism should be improved (graphics, textures, sound and lighting) so that the application will approach environments similar to those of popular commercial video games. Finally, secondary factors that contribute to enhance the learners' motivation should be isolated and studied separately, like mystery, and imagination.

7. ACKNOWLEDGMENTS

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8. REFERENCES

[1] Ahdell, R. & Andresen, G. 2001. Games and simulations in workplace elearning: How to align eLearning content with learner needs. Master of Science Thesis, Norwegian University of Science and Technology.

[2] Amory, A., Naicker, K., Vincent, J., and Claudia, A. 1998. Computer Games as a Learning Resource. In Proceedings of ED-MEDIA, ED-TELECOM 98, World Conference on Education Multimedia and Educational Telecommunications. Vol. 1, 50-55.

[3] Aschersonn N. 2000. Editorial, Public Archaeology, 1(1). 2.

[4] Asgari, M., and Kaufman., D. 2004. "Relationships among computer games, fantasy, and learning." 2nd International Conference on Imagination in Education. (Vancouver, Canada, 2004). Retrieved November 30, 2009, from http://www.ierg.net/confs/2004/Proceedings/Asgari_Kaufman.pdf.

[5] BECTA. 2002. British Educational Communications and Technology Agency: http://www.becta.org.uk/research/research.cfm?section=1&id=519

[6] Berlyne, D. E. 1960. Conflict, arousal, and curiosity. New York: McGraw-Hill.

[7] Boyle, T. 1997. Design for Multimedia learning. London, Prentice Hall.

[8] Bricken, M., and Byrne, C. 1993. Summer Students in Virtual Reality: A Pilot Study on Educational Applications of Virtual Reality Technology. In A. Wexelblat (Ed.), Virtual Reality: Applications and Explorations. CA: Academic Press, San Diego, 199-217.

[9] Brody, H. 1993. Video Games that Teach?. Technology Review, 96(8), 51-57.

[10] Brown, D.J., Cobb, S.V. & Eastgate, RM. 1995. Learning in Virtual Environments (LIVE). Virtual Reality Applications, R.A. Earnshaw, J.A. Vince & H. Jones (eds), Academic Press pp. 245-252.

[11] Caskey, J.L. 1968. Lerna in the Early Bronze Age. American Journal of Archaeology.72, 313–316.

[12] Ciavarro, C. 2006. The Design, Development and assessment of an educational sports-acrion video game: implicity changing player behavior. Master Thesis, Simon Fraser University,.

[13] Cordova D. I., and Lepper M. R. 1996. Intrinsic motivation and the process of learning: Beneficial effects of contextualization, personalization, and choice. Journal of Educational Psychology, 88, 715-730.

[14] Darken, R.P., and Sibert J.L. 1996. Wayfinding strategies and behaviours in large virtual worlds. In Proceedings of the CHI '96 conference on Human Factors in Computing Systems (Vancouver, Canada, 1996). New York: Association for Computing Machinery.

[15] Deci, E., and Ryan, M. 1985. Intrinsic motivation and self-determination in human behavior. Plenum, New York.

[16] Dimaraki, E. 2007. Ψηφιακή διαμεσολάβηση της μάθησης για το παρελθόν: σχεδιασμός εφαρμογών για την ενορχήστρωση μαθησιακής δραστηριότητας. In Νικονάνου Ν, Κασβίκης Κ., Εκπαιδευτικά ταξίδια στο χρόνο, Εμπειρίες και ερμηνείες του παρελθόντος, Πατάκης, Αθήνα.

[17] Driscoll, M. P. 1994. Psychology of learning for instruction, Needham Heights, MA: Allyn and Bacon.

[18] Duffy, T.M., and Jonassen D.H. 1992. Constructivism and the technology of instruction. Lawrence Erlbaum Associates Publishers: New Jersey.

[19] Economou, D. 2001. The role of Virtual Actors in Collaborative Virtual Environments for Learning, Ph.D. thesis. Department of Computing and Mathematics, Manchester Metropolitan University, Manchester.

[20] Facer, K. 2003. Computer games and learning. Retrieved 15 December 2009 from Future Lab, innovation in education: http://www2.futurelab.org.uk/resources/documents/discussion_papers/Computer_Games_and_Learning_discpaper.pdf

[21] Garris,R., Ahlers,R., and Driskell,.J.E. 2002. Games, motivation and learning, Simulation and gaming, An Interdisciplinary. Journal of Theory, Practice and Research. 33(4), 441-467.

[22] Gigante, M.A. 1993. Virtual Reality: definitions, history and applications. Virtual Reality Systems, R.A. Earnshaw, M.A. Gigante & H. Jones (eds.), London: Academic Press, ISBN 0-12-22-77-48-1, pp. 3-14.

[23] Hagg, R., and Konsola, D. 1986. Early Helladic architecture and urbanization. In Proceedings of a seminar held the Swedish Institute (Athens, June 8, 1985). SIMA, 76. Goteborg.

[24] Hamilakis, Y. 2004. Archaeology and the politics of pedagogy, World Archaeology, 36(2), 287–309.

[25] Inkpen, K., Upitis, R., Klawe, M., Hsu, D., Leroux, S., Lawry, J., Anderson, A., Ndunda, M., and Sedighian, K. 1994. We Have Never Forgetful Flowers in Our Garden: Girls' Responses to Electronic Games. Journal of Computers in Mathematics and Science Teaching, 13(4), 383-403.

[26] Inkpen, K., Upitis, R., Klawe, M., Hsu, D., Leroux, S., Lawry, J., Anderson, A., Ndunda, M., and Sedighian, K. 1994. We Have Never Forgetful Flowers in Our Garden: Girls' Responses to Electronic Games. Journal of Computers in Mathematics and Science Teaching, 13(4), 383-403.

[27] Jameson, J. 2004. Public Archaeology in the United States. In N. Merriman, Public Archaeology, Routledge, New York, 21- 58.

[28] Jonassen, D. 1994. Thinking technology: Toward a constructivist design model. Journal of Educational Technology, 34 (2), 34-37.

[29] Jonassen, D.H., Campbell, J.P., and Davidson, M.E. 1994. Learning with media: Restructuring the debate. Journal of Educational Technology Research and Development, 39(3), 5-14.

[30] Kagan, J. 1972. Motives and development. Journal of Personality and Social Psychology. 22, 51-66.

[31] Kaur, K. 1998. Designing virtual environments for usability. PhD thesis. Centre for HCI Design, City University London.

[32] Kernan, M. C., and Lord, R. G. 1990. Effects of valence, expectancies, and goal-performance discrepancies in single and multiple goal environments. Journal of Applied Psychology, 75, 194-203.

[33] Kirriemuir, J. 2002. Video gaming, education and digital learning technologies. D-Lib Magazine, 8(2).
[34] Kirriemuir, J., and McFarlane, A. 2004. Literature review in games and learning: A report for NESTA Futurelab, Retrieved 30 November 2009 from Future Lab, innovation in education: http://www.futurelab.org.uk/resources/documents/lit_reviews/Games_Review.pdf
[35] Kissock, J. 1987. Archaeology and its place in the primary school curriculum. Archaeological Review from Cambridge, 6(1987), 119 – 28.
[36] Klawe, M. 1999. Computer games, education and interfaces: The E-GEMS project. In Proceedings of the Graphics Interface 1999 Conference. Ontario, Canada, 36-39.
[37] Laurel, B. 1993. Computers as Theatre, Addison-Wesley.
[38] Liakos, A. 2007. Πως το παρελθόν γίνεται ιστορία;. Πόλις, Αθήνα.
[39] Locke, E. A., and Latham, G. P. 1990. A theory of goal setting and task performance, Englewood Cliffs, NJ: Prentice Hall.
[40] Malone, T. W., and Lepper, M. R. 1987. Making Learning Fun: A Taxonomy of Intrinsic Motivations for Learning. In R. E. Snow and M. J. Farr (Eds.), Aptitute, Learning and Instruction: III. Conative and affective process analyses. Hilsdale, NJ: Erlbaum.
[41] Malone, T.W. 1980. What makes things fun to learn? A study of intrinsically motivating computer games. Palo Alto Research Center, Xerox.
[42] Mantovani, G. 1996. New Communication Environments From Everyday to Virtual, Taylor & Francis.
[43] McLellan, H. 1996. Virtual realities. In D. H. Jonassen (Ed.), Handbook of research for educational communications and technology, Macmillan Library Reference, New York, USA, 457–487.
[44] McLellan, H. 2003. Virtual realities. In D. H. Jonassen and P. Harris (Eds.), Handbook of research for educational communications and technology (2nd ed), Lawrence Erlbaum Associates, New York, USA, 461–498.
[45] McManamon, F. 2000. Public education. A part of archaeological professionalism. In Smardz and Sh. J. Smith (eds.), The archaeology education handbook. Sharing the past with kids, Altamira Press and Society for American archaeology: Walnut Creek, 17-24.
[46] Mitchell, W.L. and Economou, D. 1999. Understanding Context and Medium in the development of Educational Virtual Environments. In Proceedings of the Workshop on User Centered Design and Implementation of Virtual Environments, University of York, 109-115.
[47] Neale, D. C., and Carroll, J. M. 1997. The role of metaphors in user interface design. In Helander, M. G., Landauer, T. K., and Prabhu, P., editors, Handbook of Human-Computer Interaction, 2nd edition. Elsevier Science.
[48] Pantelidis, V. S. 1991–2007. Virtual reality and education: Information sources; a bibliography. Retrieved November 10, 2007, from Virtual Reality and Education Laboratory, Department of Library Science
College of Education, East Carolina University, Greenville, NC USA: http://vr.coe.ecu.edu/vpbib.Html
[49] Pantelidis, V. S. 1995. Reasons to use virtual reality in education. VR in the Schools, 1(1), 9. Retrieved September 29, 2007, from Virtual Reality and Education Laboratory, Department of Library Science
College of Education, East Carolina University, Greenville, NC USA: http://vr.coe.ecu.edu/reas.html
[50] Papert, S. 1993. The Children's Machine: Rethinking School in the Age of the Computers. Basic Books, New York.
[51] Parker, L. E., and Lepper, M. R. 1992. Effects of fantasy context on children's learning and motivation: Making learning more fun. Journal of Personality and Social Psychology, 62, 625-633.
[52] Pivec, M., Dziabenko, O., and Schinnerl, I. 2003. Aspects of game-based learning. In Proceedings of International Conference on Knowledge Management (Austria, Gratz, July 2-4, 2003). I-KNOW '03. 216-225.
[53] Prensky, M. 2000. Digital Game-Based Learning. New York, McGraw Hill.
[54] Prensky, M. 2002. The motivation of gameplay. On the Horizon, 10(1).
[55] Rogers, A. 1999. Η εκπαίδευση ενηλίκων. Μεταίχμιο, Αθήνα.
[56] Salomon, G., Perkins, D., and Globerson, T. 1991. Partners in cognition: Extending human intelligence with intelligent technologies. Educational Researcher.
[57] Sanders, D.H. 2000. Archaeological publication using virtual reality: case studies and cavets. Virtual Reality in Archaeology, J.A. Barcelo, M. Forte and D. Sanders (eds.), ArchaeoPress, Oxford, 2000, (British Archaeological Reports International Series S843 2000), ISBN: 1 84171 047 4, pp. 37-46.
[58] Scardamalia, M., Bereiter, C., McLean. R.S., Swallow, J., and Woodruff, E. 1989. Computer-supported intentional learning environments. Journal of Educational Computing Research, 5(1), 51-68.
[59] Schadla- Hall, T. 1999. Editorial: Public Archaeology. European Journal of Archaeology., 2(2), 47-158.
[60] Sedighian, K. 1997. Challenge-Driven Learning: A Model for Children's Multimedia Mathematics Learning Environments. Conference of Educational Multimedia and Hypermedia and Educational Telecommunications, 1997, Calgary, Canada.
[61] Slater, M., and Wilbur, S. 1995. Through the looking glass world of presence: FIVE: a Framework for Immersive Virtual Environments. In Proceedings of FIVE '95: Framework for Immersive Virtual Environments. (London, 1995). University of London.
[62] Smith, K.C. 1991. At Last a Meeting of the Minds. Archaeology, 50(1), 36-46, 80.
[63] Virvou, M., Katsionis, G., and Manos, K. 2005. Combining Software Games with Education: Evaluation of its Educational Effectiveness. Educational Technology and Society. Journal of International Forum of Educational Technology and Society and IEEE Learning Technology Task Force, 8(2).
[64] Virvou, M., Manos, K., Katsionis, G., and Tourtoglou, K. 2002. Incorporating the Culture of Virtual Reality Games into Educational Software via an Authoring Tool. In Proceedings of the IEEE International Conference on Systems, Man and Cybernetics (SMC 2002), Tunisia, 422-428.
[65] Weiberg, E. 2007. Thinking the Bronze Age, Life and Death in Early Helladic Greece, Uppsala Univertitet.
[66] Wilcoxon, F. 1945. Indivindual Comparisons by ranking methods. Journal of Biometrics Bulletin, 1(6), 80-83.
[67] Winn, W., 1993. A conceptual basis for educational applications of virtual reality. Retrieved 10 February 2010 from Human Interface Technology Laboratory: http://www.hitl.washington.edu/publications/r-93-9/.
[68] Youngblut, C. 1998. Educational uses of virtual reality technology. Alexandria, A: Institute for Defense Analyses.
[69] Γιαλαμάς, Β. 2005. Στατιστικές τεχνικές και εφαρμογές στις επιστήμες της αγωγής. Εκδόσεις Πατάκη, Αθήνα.
[70] Ρούσσας, Γ. 1994. Στατιστική συμπερασματολογία: Έλεγχος υποθέσεων. Εκδόσεις Ζήτη, Θεσσαλονίκη.